Discover Environment

Review

Interdisciplinary strategies for the management of harmful algal blooms: prospects & comprehensive review

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Abstract
In recent decades, with the continuous development of global climate change and industrialization, the majority of coastal waters globally are experiencing escalating eutrophication, with a concomitant increase in the frequency of harmful algal blooms. The large-scale reproduction or decay of algae will reduce the content of dissolved oxygen in the water, leading to the death of a large number of aquatic animals,and seriously damaging the water ecosystem. Moreover, the algal toxins released by harmful algae can not be ignored. Some algae toxins have obvious and harmful effects even though their cellular concentrations are small. The algae toxins produced by harmful algae are constantly enriched through the food web, ultimately endangering human health.At present, the algae bloom treatment methods are roughly divided into three categories: physical control methods, chemical control methods and biological control methods. However, in governance, there is always a deficiency in a single method, and the comprehensive control strategy combining physical, chemical and biological methods can make up for the deficiency of a single method. This review provides a comprehensive analysis of the latest developments in global harmful algal blooms treatment technologies, assesses the adverse impacts of harmful algal blooms on the environment and human health, and proposes viable integrated management strategies for future harmful algal blooms. We summarize the application of physical, chemical, and biological methods in the treatment of algal blooms and highlight the significance of an interdisciplinary approach in the future management of harmful algal blooms.

Keyword Cyanobacteria bloom · Resource utilization · Green governance · Review

1 Introduction

Harmful algal blooms (HABs), characterized by the abrupt and excessive proliferation of algae, primarily result from the eutrophication of aquatic ecosystems. The frequency of such events has notably increased in recent decades, precipitating significant ecological and economic detriments, including the extensive mortality of aquatic organisms and substantial financial losses within the fisheries and aquaculture sectors. A poignant illustration of this issue occurred in the autumn of 2021 when *H. akashiwo* blooms resulted in the decimation of four tons of farmed salmon across several

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aquaculture facilities in the Koma fjord, Los Lagos, incurring losses estimated at $121,000 [1]. The genesis of HABs is intrinsically linked to eutrophic conditions and exacerbated by global climate dynamics, where elevated nutrient levels facilitate algal reproduction, and climate change amplifies algae's competitive dominance. John et al.[2] underscores the pivotal role of nitrogen and phosphorous concentrations in fostering algal proliferation, with critical thresholds delineated at 0.5 mg·L^{-1} for total nitrogen (TN) and 0.02 mg·L^{-1} for total phosphorus (TP). Further, Paerl et al. [3] have elucidated the impact of climate change on HABs, highlighting how the synergistic effects of global warming and extreme weather events enhance algae's environmental resilience and dispersion due to their high adaptability [4].

As shown in Fig. 1, the optimal strategies for HABs prioritize the mitigation of eutrophication and the reduction of algal biomass within aquatic ecosystems, aiming to curtail bloom proliferation at minimal economic cost. Contemporary appreciable strategies to HABs remediation are categorized into physical, chemical, and biological methodologies. Physical techniques are employed to directly modulate nutrient concentrations and algal biomass, thereby inhibiting bloom formation. Chemical strategies leverage reagents to diminish nutrient levels and detoxify waters, attenuating the severity of blooms. Biological interventions utilize the trophic dynamics among flora, fauna, and microorganisms to regulate nutrient availability, thus fostering water quality improvement. Notwithstanding, the application of singular remediation techniques frequently encounters constraints regarding efficacy and sustainability, seldom achieving long-term, impact bloom control.

Accordingly, the management of HABs necessitates a departure from reliance on isolated methodologies towards the adoption of an integrated, multidisciplinary treatment paradigm. This holistic approach entails the synergistic application of physical, chemical, and biological techniques, harmonizing their respective advantages to formulate a cost-effective and efficacious bloom control integrated strategy. Such a comprehensive management framework underscores the importance of interdisciplinary collaboration, amalgamating diverse preventive and remedial measures to ensure the sustainable governance of HABs phenomena.

As illustrated in Fig. 2, this manuscript elucidates the etiological factors and ecological detriments associated with HABs, alongside elucidating the intricacies of contemporary remediation challenges (Sect. 2). It proceeds to critically evaluate the efficacy of extant treatment modalities, encompassing physical (Sect. 3), chemical (Sect. 4), and biological (Sect. 5) strategies, in the context of HABs management. Central to this discourse is the articulation of the imperative for an integrative control paradigm, detailing the conceptual framework underpinning a comprehensive treatment strategy (Sect. 6). Subsequent deliberations (Sect. 7) are dedicated to prognosticating the evolution of holistic algal remediation techniques, anticipating advancements in the interdisciplinary coordination of treatment methodologies. The concluding section (Sect. 8) accentuates the criticality of a multidisciplinary approach towards the efficacious governance of HABs phenomena, advocating for an enhanced collaborative engagement among scholars across diverse scientific domains. This review emphasizes the necessity for concerted research efforts aimed at refining and augmenting integrated HABs management strategies, thereby ensuring the ecological integrity and sustainability of aquatic ecosystems.

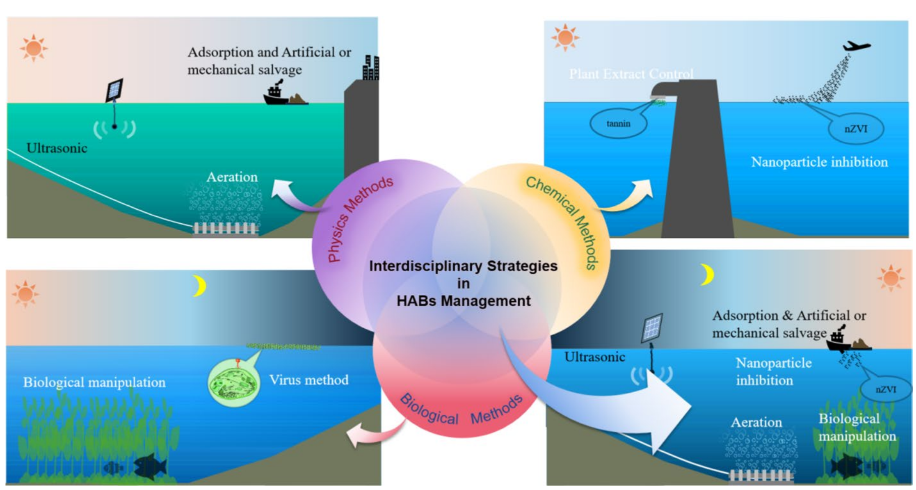

Fig. 1 The appreciable strategies for HABs management

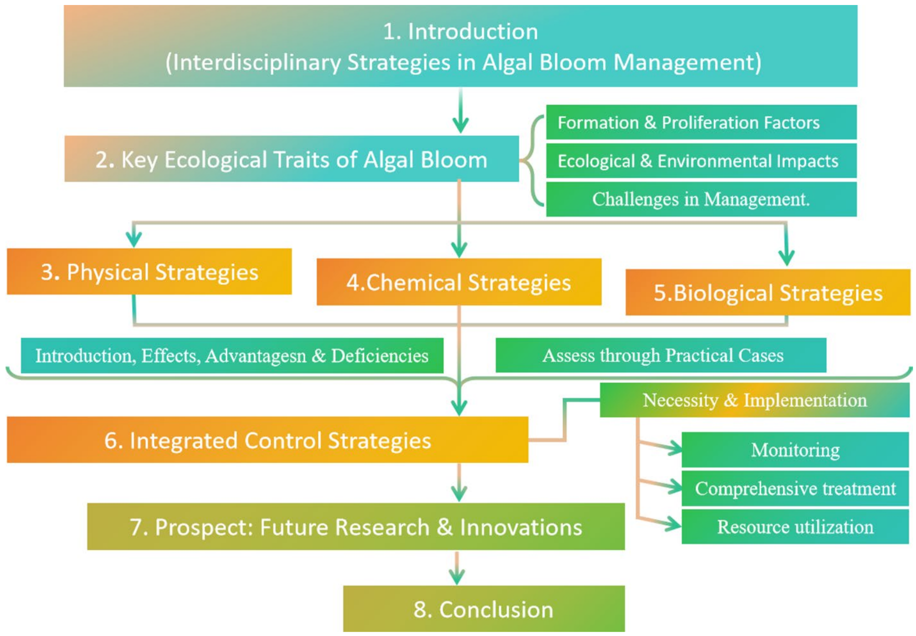

Fig. 2 Outline of this review on Integrated Strategies in HABs Management

2 Key ecological traits of HABs

2.1 Factors contributing to the formation and proliferation of HABs

The genesis and escalation of HABs are attributed to several factors, including nutrient enrichment, thermal preferences, solar irradiance, and alterations in water chemistry, such as pH and turbidity. These elements coalesce into two primary drivers: eutrophication of the coastal aquatic environments and climatic alterations. Eutrophic conditions provide a fertile arena for the unchecked replication of algae, leading to significant aggregations that heighten water turbidity, thereby obstructing subaquatic light penetration [5]. This limitation on light availability hampers the growth of other photosynthetic organisms, including large benthic flora, culminating in the degradation of aquatic biodiversity and the establishment of bloom conditions.

Concurrently, the role of global climate dynamics in fostering HABs is increasingly substantiated, with climatic variability significantly reshaping coastal aquatic ecosystem characteristics [6]. For instance, rising global temperatures are correlated with enhanced precipitation events, escalating the influx of nutrients into coastal aquatic systems, thus intensifying eutrophication and favoring cyanobacterial proliferation. Moreover, climate-induced extreme weather events, exemplified by Hurricane Katrina [7] and the floods in Pakistan [8], exert profound impacts on coastal aquatic ecosystems, deteriorating the habitat suitability for diverse aquatic life. Nonetheless, certain algal species, notably cyanobacteria [9], exhibit adaptive advantages such as buoyancy regulation and efficient nutrient uptake mechanisms (such as Carbon Concentration Mechanisms (CCM) and phosphorus acquisition strategies), granting them a competitive edge under adverse environmental conditions [10]. In the case of *Microcystis* as an example. In the review by Song et al. [11] suggesting that the diversity of phosphorus sources in aquatic environments leads to greatly different availability of organisms in aquatic ecosystems, and *Microcystis* have a strong ability to use phosphorus, other metabolic processes can be mobilized to reduce energy expenditure when phosphorus sources are scarce. At the same time, phosphorus deficiency can promote the transformation of organic P by *Microcystis*. This utilization ability allows *Microcystis* to resist disadvantages from aquatic ecosystems, reflecting a strong competitive advantage. These adaptations facilitate their dominance in nutrient-rich, variable environments, precipitating the emergence and persistence of HABs [5].

2.2 Ecological and environmental impacts of HABs

The reciprocal interactions between HABs, climatic warming, and aquatic eutrophication are manifestly evidenced by bidirectional feedback mechanisms. As shown in Fig. 3, the senescence and sedimentation of algal biomass instigate a cascade of biogeochemical processes in the benthic layers, where bacterial degradation of this organic

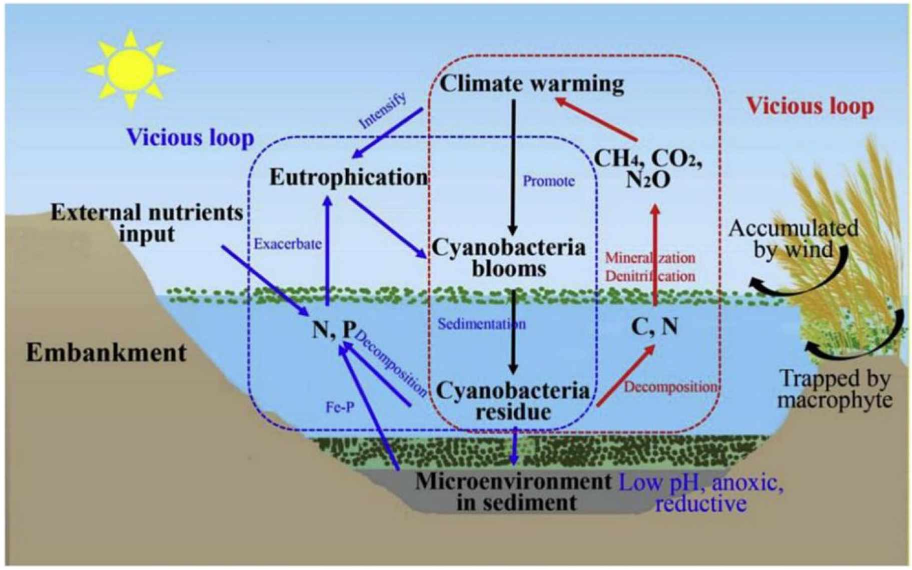

Fig. 3 Conceptual diagram of the interplay between climatic warming, aquatic eutrophication, and HABs[12]

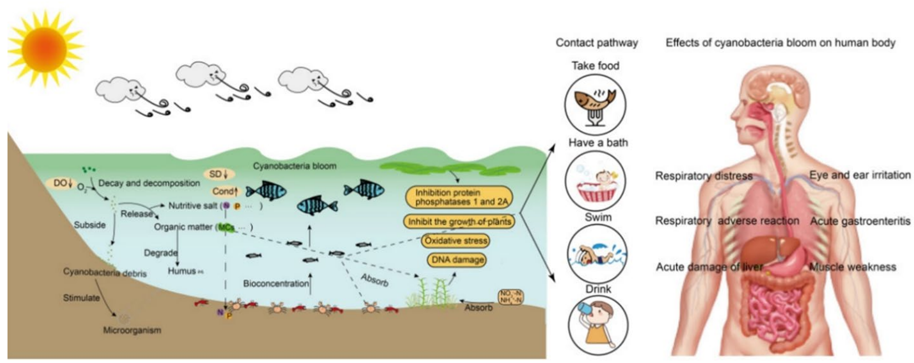

Fig. 4 Main harm of algae bloom to water body, aquatic life and human body[14]. *DO* dissolved oxygen, *SD* water transparency, *Cond* conductivity, *MCs* microcystins

matter leads to substantial oxygen consumption, thereby precipitating hypoxic conditions [12]. The decomposition of algal biomass further results in the release of carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and other greenhouse gases, exacerbating the greenhouse effect and, consequently, global warming. Concurrently, the decay process liberates significant quantities of nitrogen and phosphorus, augmenting the nutrient load in the water column, thus perpetuating a cycle of eutrophication and promoting further algal proliferation. This intricate web of interactions underscores the existence of self-reinforcing cycles that exacerbate the severity of HABs, eutrophication, and climate change in a synergistic fashion [3].

HABs precipitate a rapid depletion of dissolved oxygen (DO) levels, engendering hypoxic or anoxic conditions detrimental to aquatic life. The resultant biodiversity loss, coupled with alterations in microbial community composition and function, signifies a profound disruption of coastal aquatic ecosystem integrity [13]. As shown in Fig. 4, the nutrient enrichment from nitrogen and phosphorus not only compromises water transparency but also catalyzes a further decline in DO concentrations [14]. Post-mortem algal biomass undergoes decomposition, contributing to a surge in soluble organic matter that transitions into humic substances[15], thereby facilitating the enhanced cycling and bioavailability of nitrogen and phosphorus, and exacerbating eutrophic conditions. Additionally, metabolic byproducts of HABs can introduce odors, increase turbidity, and compromise the potability of water supplies [16], with ramifications extending to the bioaccumulation of toxins within aquatic food webs [17]. This accumulation poses significant health risks to terrestrial and aquatic fauna, as well as to humans, as evidenced by historical incidents of morbidity and mortality attributed to cyanotoxin exposure [18]. For example, in 1996, 50 dialysis clinic patients died of using water contaminated with cyanobacteria blooms [19]. And in 2020, at least 330 prairie elephants in Africa died of excessive cyanobacteria toxins [20]. Notably, epidemiological studies have correlated exposure to surface waters affected by *Microcystis* blooms with elevated incidences of hepatocellular carcinoma, further underscoring the public health implications of HABs events [17].

2.3 Challenges in the management of HABs

At present, algae bloom treatment methods are emerging in an endless stream, but there are still many challenges in the management of algae blooms.

2.3.1 Diversity of the factors responsible for HABs

As mentioned above, there are many factors affecting algae blooms, and the problems caused by algae blooms are also very diverse. Figure 5 illustrates the multiple interacting environmental variables that control the formation of algae blooms in freshwater bodies. It shows key factors controlling algae growth and dominance, including nutrient input, water transparency, mixing conditions, water residence time, temperature, and predation [3].

This complexity is analogous to the multifactorial nature of natural disasters, necessitating sophisticated management strategies to mitigate the adverse impacts of HABs on both economic stability and public health, particularly in coastal regions [21]. The challenge lies in devising cost-effective and efficient real-time monitoring systems to preemptively address the fluctuating dynamics of aquatic environments prone to HABs incidents.

2.3.2 Imperfections in governance approaches

Existing governance methods, whether physical, chemical or biological, have their own shortcomings, and some will even lead to more serious environmental hazards. Like the aeration method in physical control method, Balangoda et al.[22] showed that the active phosphorus content of water body after aeration was 3 ~ 8% higher than that of non-aeration water, which showed that aeration method would decompose the organic matter in the sediment, increase the degree of water eutrophication and aggravate the outbreak of algae. This problem also exists in chemical control method. Zhou et al. studied the method of using H_2O_2 to inhibit algae bloom, and found that H_2O_2 inhibited algae bloom for a very short time, which easily led to the recovery or even rebound of algae biomass [23]. So the treatment methods of algae blooms need to be improved.

2.3.3 Deficiencies in management experience

As early as 150 years ago, George Francis[24] focused on the problems of cyanobacteria blooms in freshwater. Despite the increasing incidence and severity of blooms over the last five decades, exemplified by species such as *Microcystis*, fish algae, and silk algae, the collective experience in effective bloom management remains insufficient [25]. Notable incidents, such as the 2014 Toledo water crisis affecting over 100,000 residents [26], and the 2007 Taihu Lake cyanobacterial outbreak in China [27], impacting the drinking water supply for millions. All these evidences illustrated the global shortfall in adept HABs governance [28]. It is a critical need for the advancement of knowledge and practices in the management of HABs to forestall future ecological and public health crises.

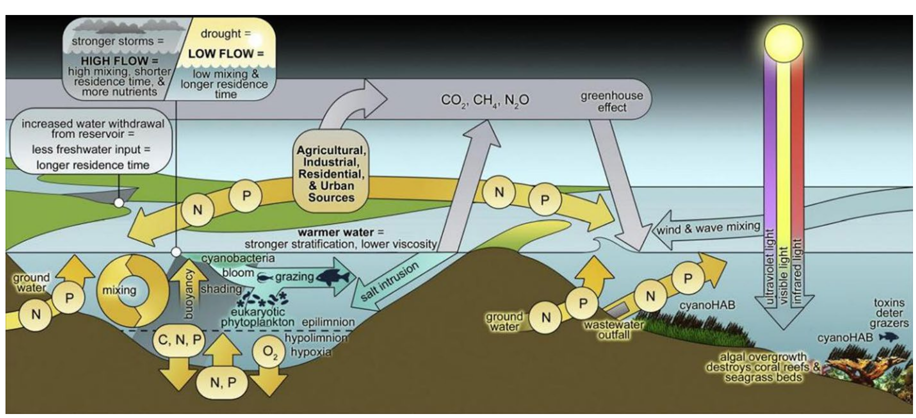

Fig. 5 Interactions of HABs with aquatic ecosystems and key factors controlling algal growth and dominance [3]

3 Physical strategies in HABs management

The deployment of physical methodologies for the management of HABs entails the direct manipulation of algal biomass within aquatic environments, aiming to curtail the proliferation of HABs. These strategies offer rapid, albeit temporary, solutions for bloom suppression. Principal among these techniques are dredging, water column mixing, aeration, ultrasonic interventions, and mechanical or manual algae removal. Each approach is characterized by its unique set of benefits and limitations, necessitating a nuanced understanding for effective application.

3.1 Overview of physical control strategies and their efficacy

As shown in Table 1, we have summarized the methods of physical control strategies of HABs in recent years, as well as their advantages and deficiencies. These physical approaches, while instrumental in offering immediate relief from HABs, must be integrated within a broader management strategy that addresses the root causes of eutrophication and algal proliferation, ensuring sustainable aquatic ecosystem health.

3.2 practical cases of physical treatment of HABs

Yan et al. [29] provide a comprehensive review of three decades of efforts in aquatic pollution management within Taihu Lake, highlighting significant sediment dredging activities initiated in 2007 that culminated in the removal of over 10 million tons of cyanobacteria (wet weight). This intervention led to a notable reduction in the nitrogen levels within the lake. Despite these efforts not yielding substantial improvements in water clarity or quality, the application of physical methodologies in mitigating algal proliferation demonstrates a tangible, albeit limited impact on controlling HABs phenomena.

4 Chemical strategies for HABs management

4.1 Overview of chemical strategies and their efficacy

As shown in Table 2, chemical treatment method refers to the use of chemical principles, through the reaction of chemical reagents and pollutants to reduce the content of nutrient salts in the water, and remove the water may contain toxins in cyanobacteria, so as to weaken the bloom phenomenon[38]. The chemical degradation methods usually used are using algal killer (nitrite, H_2O_2) [39], metal salt flocculant (mainly aluminum salt and iron salt) [40], natural compounds and their extracts [41], nanomaterials [42], and other reagents(photo-encyclogenic material decomposition method) [43] with the effect of algae killing or algae suppression for degradation. At the same time, there are also certain requirements for the treatment personnel. Chemical degradation methods underscore the rapid amelioration of water quality afforded by chemical treatments, and highlighting their cost-effectiveness as a key advantage [39].

Nonetheless, the application of chemical treatments necessitates meticulous control over the type and dosage of algicidal agents to prevent secondary contamination during remediation processes. Furthermore, the requisite expertise of operational personnel is paramount, given the dynamic and often unpredictable conditions of aquatic ecosystems. This is particularly challenging for non-specialist individuals, such as untrained farmers, who may struggle to ascertain optimal dosages. Under-dosage fails to curb cyanobacterial growth effectively, whereas over-dosage risks the collateral poisoning of non-target aquatic fauna.

4.2 Practical cases of chemical treatment of HABs

Kibuye et al. [44] present an insightful evaluation of a chemical intervention in a 64-ha reservoir, employing 4 ~ 15 kg ha^{-1} of liquid $Fe(SO_4)_3$ with application frequencies of 5 ~ 6 times annually. This approach effectively reduced phosphorus concentrations, curtailed phytoplankton proliferation, and prevented algal enrichment in subsequent years, demonstrating the potential of chemical strategies in sustainable HABs management. Further, in the northwest regions of Taihu

Table 1 The effects, advantages and deficiencies of physical control strategies

Governance approach	Method introduction	Advantages	Deficiencies	References
Physical control strategies				
Dredging method	As a common geoengineering method for controlling nutrient internal loads, environmental dredging has been used in many lake restoration projects. It can directly remove the sediment that contains a large amount of nutrients, thus controlling the content of nutrient salts in water	It can immediately reduce the eutrophication degree of water bodies	It has no obvious effect on long-term algal treatment and will cause harm to water bodies	[30]
Mixed method	The mixing method is the mechanical operation of the water cycle within the lake or reservoir to weaken or eliminate the density stratification of the water column.It can mix the water column evenly, leading to the near-isothermal state of the eutrophic lake, increasing the depth of the mixing layer and the dissolved oxygen content	The increase of dissolved oxygen can be inhibited to achieve the purpose of controlling the nutrient content in the water body, and weakens the access to light and nutrients by algal	Equipment purchase and maintenance costs are high, and mixing can lead to resuspension of nutrients, which may amplify algae growth and lead to failure of algae control	[31]
Aeration method	Aeration improves the concentration of dissolved oxygen in water by mixing water and gas, suppresses the release of nutrients and ions in water in a short time, and changes the spatial and temporal distribution pattern of algae to reduce the large accumulation of algae on the water surface	The cost is low, the application is broad, and the reproduction of algae can be better controlled	The use of aeration in some waters with high levels of endogenous eutrophication allows nutrients from sediments to return to the water body	[32]
Ultrasonic	Ultrasonic wave in water will cause a series of compression and sparse cycle, leading to a large number of vacuoles in water, a large number of vacuolar implosion will produce a local temperature up to 5000℃, 100 MPa pressure and a large number of free radical, so as to inhibit the growth rate of algae. And the extreme environment generated by ultrasonic treatment can cause the gas vacuoles in some algae to collapse, resulting in the loss of buoyancy, and then settle to the bottom; or directly destroy the algae cell membrane, namely cell lysis, thus reducing the algae biomass in the water	It can be used on large water surfaces and is environmentally friendly. Lower cost compared to other methods	It can cause damage to other organisms in the water body, such as fish, and may return nutrients from the bottom sediment to the water body	[33, 34]
Manual or mechanical salvage method	Artificial or mechanical salvage method is the most common, the most extensive, the most mature algae bloom control method. Direct remove algae from the surface to reduce algal biomass	It is the most widespread and mature algal treatment technology available today, and the cost is low	Because this method does not address the root cause of HABs, they lead to continuous regeneration	[35]

Table 1 (continued)

Governance approach	Method introduction	Advantages	Deficiencies	References
Adsorption method	Adsorption technology is widely used to remove pollutants in aqueous solution. The adsorption mechanism is mainly affected by the surface properties of the adsorbent, the experimental conditions and the chemical properties of the target material. For example, the surface of the adsorbent is modified to make it with positive or negative charge, so as to improve the adsorption capacity, and to adsorb the algal cells in the water	It prevents the harm of cyanotoxins and can be conducive to the resource utilization of algal	Because this method does not address the root cause of HABs, they lead to continuous regeneration	[36]
Ultraviolet irradiationmethod	UV irradiation can directly destroy algal cell structures and promote the biodegradation process, thereby altering the photosynthetic and antioxidant enzyme activities within algal cells, which can lead to their death or inactivation. Furthermore, UV radiation significantly affects the coagulation effectiveness for algae	It prevents the harm of cyanotoxins and is environmentally friendly. Beneficial for the coagulation and sedimentation of algae, it is also cost-effective	In aquatic systems exhibiting elevated levels of turbidity, the photodegradation efficacy conferred by ultraviolet irradiation is potentially attenuated	[37]

Table 2 The effects, advantages and deficiencies of chemical control strategies

Governance approach	Method introduction	Advantages	Deficiencies	References
Chemical control strategies				
Algaecides	Altalcide is a chemical agent used to control the algae bloom problem, which can effectively control the growth of excessive algae in water bodies	The algae removal efficiency is high and the effect is remarkable	Toxic to the environment, and excessive use cause environment secondary harm, and long-term use requires more doses	[39]
Metal salt flocculants	The principle of metal salt flocculants is to add metal salt solution (such as aluminum salt and iron salt) to the water, so that metal ions are complexed with algae cells and organic matter in the water to form flocculation, so that algae cells and suspended matter gather together, and then settle to the bottom or be removed by filtration	Reduce the content of nitrogen and phosphorus nutrients in water bodies and improve water quality	The metal salt flocculation method alone can only reduce the degree of nutrient in the water, but does not reduce the content of endogenous nutrients	[40]
Nanoparticle inhibition	In terms of algae bloom management, nanoparticles show their potential. NZVI (zero-valent iron nanoparticles) can control algae bloom through cell lysis, and then nZVI is converted into non-toxic $Fe\ (OH)_3$ to produce flocculent with phosphate and cell debris in water, helping to control HABs	It prevents the harm of algal toxins, purifies water quality, and controls the degree of eutrophication in the water body	Suitable for small water areas without reducing the endogenous nutrient content	[42, 46]
Plant extract control method	Many plants are natural algicide producers, such as seaweed, rice, and even walnuts and roses, and the extracts of these plants can inhibit algae blooms through allelopathy	Natural extract,easy to prepare	It may have a negative impact on seedling germination and growth	[41]
Photo-encyclogenic material decomposition method	The principle of photocatalytic materials decompose algae and toxins is to use the photocatalytic oxidation capacity of photocatalytic materials (such as TiO_2)	Low energy consumption and simple method	The treatment efficiency is low, and need to develop new materials	[43]

Lake, Zhao et al.[45] assessed the efficacy of eucalyptus-based systems for algal control. The investigation revealed that the eucalyptus planting system and eucalyptus leaf extract system achieved inhibition efficiencies of 85.8% and 20.9%, respectively, underscoring the superiority of certain chemical methodologies in mitigating HABs.

5 Biological strategies in HABs management

5.1 Overview of biological strategies and their efficacy

Biological control in the context of HABs management entails the strategic utilization of flora and fauna to modulate nutrient concentrations within aquatic ecosystems via trophic interactions, or the deployment of microorganisms to induce algal cell lysis and biological flocculation, thereby reducing algal cell densities [47].

As shown in Table 3, we have summarized the methods of biological control strategies of HABs in recent years, as well as their advantages and deficiencies. This method hinges on the principles of predation, competitive exclusion, and microbial degradation[48] as mechanisms to suppress algal proliferation. Such biological interventions[49] offer a symbiotic approach to bloom management, leveraging natural ecosystem dynamics for algal control. However, the effectiveness of these strategies can be contingent upon specific environmental conditions and the complex interplay between various biological agents, necessitating careful planning and execution for successful bloom mitigation.

5.2 Practical cases of biological treatment of HABs

Liu et al. [50] executed an innovative biological manipulation endeavor within West Lake, Huizhou, China, where, despite significant reductions in nutrient loads, turbidity issues persisted. By substituting benthic fish species with planktivorous fish and introducing extensive underwater vegetation, they successfully revitalized the eutrophic shallow lake. This intervention, as documented in "Successful restoration of a tropical shallow eutrophic lake: Strong bottom-up but weak top-down effects recorded" (Fig. 6), exemplifies the efficacy of novel biological engineering approaches in resolving HABs challenges. Concurrently, researches also explored the use of bivalve organisms [51] and filter-feeding fish [52] as a biological treatment for HABs, discovering that an increase in the biomass of these filter feeders corresponded with a reduction in algal concentrations. This outcome indicates that predator-induced top-down control exerts a more substantial impact on algal populations than the nutrient-driven bottom-up effects within the scope of biological manipulation strategies.

6 Integrated strategies

6.1 The necessity of comprehensive treatment of HABs

In the short-term prevention and control, physical control and chemical control are undoubtedly the most effective methods. These two methods can quickly reduce the algae biomass in water, and have significant results in short-term control. However, for the chemical control method, the amount of chemical reagents is the key. If the dose is not controlled, the burden increases the ecosystem and causes secondary damage. For the physical control method, the treatment efficiency cannot reach the daily clearance at low cost. In the long-term control, biological control is the best choice. The use of biological treatment method to control algae blooms can be controlled by ecological means such as food webs, or directly killed by viruses and other organisms, but there are problems such as how to prevent and control the biological invasion caused by the introduction of organisms. In addition, in some waters with high algae bloom degree, the effect of biological control method is not obvious or the cost is too high, and there is no obvious effect on short-term control [9]. Therefore, we need a new comprehensive management strategy, which can integrate the advantages of each method and combine a variety of treatment methods to jointly control algae blooms.

Table 3 The effects, advantages and deficiencies of biological control strategies

Governance approach	Method introduction	Advantages	Deficiencies	References
Biological control strategie				
Bacteriologic control	HABs can be controlled by introducing other species of bacteria, such as *Myxococcus* or *Streptobacteria*, which can kill algae by having secreted extracellular products, contacting lysis, and wrapping lysis	It can inhibit the photosynthesis and metabolism of algal, thereby controlling the growth of algal, and has strong selectivity	The effect is not obvious in the waters with deep HABs, and the treatment time is long	[48] [53]
Biomanipulation	Biological manipulation refers to the control of algae biomass in water through the food web in the water ecosystem, which is divided into two classics: Classical biomanipulation means to change the composition and quantity of predators in the ecosystem to control the biomass of algae in water; Non-classical biomanipulation method is the direct use of filter feeding fish or other organisms directly swallowed, achieving the purpose of algae bloom management	The use of biological means to control the growth of algal also controls the degree of eutrophication of water bodies	This can lead to the destruction of existing biodiversity, which is more costly in large waters	[49]
Virus law	Cyanophages, viruses that target and lyse cyanobacteria, employ both the lysogenic and lytic cycles upon infection. In the lysogenic cycle, cyanophage DNA integrates into the host algal genome, while the lytic cycle results in the direct disintegration of algal cells. Leveraging the specificity of cyanophages towards their algal hosts offers a novel, targeted approach to HABs mitigation	It is highly specific and will not harm other aquatic organisms	The algae phage culture technology is immature, and algal can also mutate, resulting in the weakening of the algae phage effect	[16] [54]
Shading technology method	Shade method is a way to control algal growth by reducing light exposure. By blocking the sunlight, it limits the light that algae need for photosynthesis	The shading technology method is an environmentally friendly and cost-effective method	It is not suitable in large open water bodies and needs to be used in combination with other algal control methods	[55]

Fig. 6 Huizhou West Lake before and after the governance of the landscape changes [50]

6.2 Implementing integrated strategies

Because the single algae bloom treatment method can not meet the expected requirements. Therefore, we propose a comprehensive treatment strategy based on three governance methods, integrating modern advanced monitoring technology and artificial intelligence technology, and implementing a comprehensive treatment strategy, to achieve low carbon, low energy consumption, low cost and high efficiency of algae bloom control purpose.

6.2.1 Monitoring of the HABs.

Real-time and rapid monitoring of harmful HABs is the basis for the prediction and control of HABs. Kibuye In his second review, site-specific characteristics, water quality parameters, and ecological impacts should be fully monitored and reported before, during, and after the use of specific treatment methods [44]. So the development of monitoring technology will help us to develop and implement control methods before the spread of algae blooms to prevent the spread [56]. At present, the advanced monitoring means include remote sensing monitoring, molecular biology monitoring, and so on.

As a tool of long-term monitoring program, remote sensing monitoring technology has macroscopic, dynamic, rapid, large-scale and periodicity. Satellite remote sensing provides a means of observing HABs with unprecedented frequency and spatial coverage [57]. Remote sensing monitoring is the relationship between the inherent optical properties of substances in the water and remote sensing reflection to monitor the composition of the substances in the water in real time [58]. Molecular biology monitoring [59] was used in water quality detection in recent years, using molecular biological methods (such as biological sensors, microarray, RT-PCR, etc.) targeted observation of algae, while processing a large number of algae samples. For example, fluorescence in situ hybridization (FISH) technology can easily use an optical microscope to identify different algae of similar morphology.

A large number of studies have shown that the microalgae near the coast of China have a biomass surge, frequent algae blooms, and variable composition of algae blooms [60], which requires a more accurate early warning system and detection technology. The combination of new monitoring technology and traditional detection methods (in situ detection), and with the assistance of artificial intelligence and big data, the establishment of a prediction model of algae blooms will contribute to the early warning of large-scale harmful algae blooms [61].

6.2.2 Comprehensive treatment of HABs

Comprehensive control of various prevention and control means can quickly prevent the expansion of algae bloom, and achieve the radical cure of algae bloom. Such as physical control and biological control of comprehensive use (Fig. 7), through remote sensing monitoring means such as algae bloom in the water, select the appropriate physical control method and biological control method, after the effect of physical control method, biological control should be used immediately to reduce the eutrophication of the water body,and expand the results of physical control to achieve the purpose of eradicating water bloom [9].

6.2.3 Resource utilization of algae blooms

The outbreak of harmful algae blooms will cause great damage to the environment, and the treated algae mud will still become a new source of pollution after treatment. However, the main metabolites of algae, such as carbohydrates, proteins and lipids in cells, can still be effectively used [36]. Therefore, looking for possible resource utilization of algae is an important research direction in the future. As shown in Table 4, we have summarized the possible resource utilization of algae. Algae stand as a multifunctional resource offering sustainable solutions across energy, environmental, and economic sectors. Their cultivation and utilization not only contribute to resource efficiency but also to combating environmental challenges such as climate change and water pollution.

7 Prospect: Future Research and Innovations

7.1 Emerging technologies and strategies

The development of novel nanomaterials is pivotal in the physical and chemical management of HABs. These materials, characterized by extensive surface areas and reactivity, enable more efficient interaction with cyanobacterial cells. Customizable to target specific cyanobacterial species, these nanomaterials can potentially minimize environmental

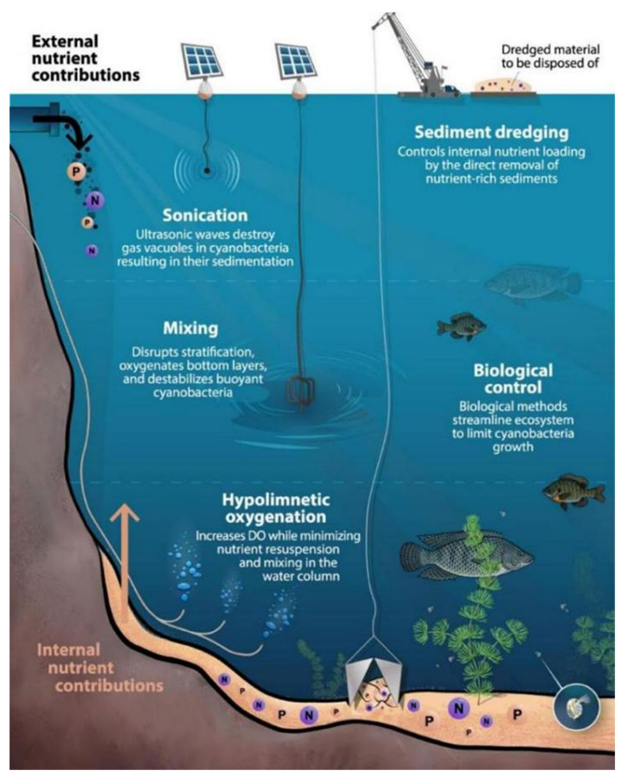

Fig. 7 The combination of biological control and physical control[9]

Table 4 The possible resource utilization of algae

Resource utilization method	Specific utilization method	References
Biodiesel	Algae can be refined into biomass diesel oil because it has the key advantage of high lipid yield compared with fuel crops such as jatropha curcas, rapeseed, and corn. In the United States (2007), algal biomass requires only 1% to 3% of the planting area to meet 50% of the transport fuel demand.The key process of microalgae for biodiesel production is to separate it from the water body, which can be better separated by the centrifugation method, but the energy consumption of the process is relatively high.Therefore, exploring the possible biodiesel production path of microalgae still needs to be further studied	[62]
Incineration and gasification for power generation	Biomass gasification power generation is a more advanced new technology, mainly through combustion, gasification, and other thermochemical processes to generate fuel.This method can effectively use the algae mud with high cyanotoxin content to generate electricity. After thermochemical processes such as incineration and gasification, the organic components can become biomass synthetic production gas (methanol and hydrogen can be separated) and inorganic residue, so that the algae mud can be safely disposed	[63]
Synthesis of chemical raw materials through genetic engineering	Algae are photoautotrophic organisms that can convert light energy into chemical energy. With this process, the carbon from the atmosphere is converted into organic compounds, which are a natural catalyst for the synthesis of chemical raw materials. For example, Consume succinate precursors and polysaccharides (succinate dehydrogenase (SdhAB) and glycogen synthase (GlgC)) in succinate-producing strains of *S. elongatus PCC 7942* by CRISPRi technology, which increased their succinate production by 82%[64]	[65] [66]
New nutrition products	Cyanobacteria contain a large number of beneficial compounds, such as carbohydrates, vitamins, carotenoids, lutein, fatty acids, and protein, that can be made into health nutrition consumption to help humans prevent diseases. A great advantage of using cyanobacteria production is its high production on non-cultivated land, which does not affect the agricultural crop	[67]
Biological feed, fertilizer	Algae contain a large number of algal proteins and fatty acids, which is a good animal husbandry feed, and can become a good source of protein in livestock feed. At the same time, it is rich in nitrogen and phosphorus, so it can be used as a good fertilizer.And the biofilm formed by cyanobacteria can effectively maintain the soil moisture. However, because it is difficult in separating the cyanobacteria toxins contained in cyanobacteria, this fertilizer is used only for ornamental plants, but not for edible crops	[68]
New carbon material: algal carbon	Algae are rich in proteins, cellulose, and heterocyclic atoms, which are suitable for making heterocyclic atoms. M. Sevilla et al.studied the generation of algal carbon by hydrothermal carbonization, affirming the ability of algae to generate heterocyclic atoms-doped hydrogen carbon (HTC) materials. Hou et al. reviewed various preparation methods for algal carbon (direct carbonization, hydrothermal carbonization, activation, etc.) and introduced the potential of algal carbon in adsorbents, chemical catalysis, electrode materials, etc	[69] [70]
Nanomaterials were made from cyanobacteria	Because of the diversity of cyanobacteria and their resistance to metal ions, cyanobacteria are not prone to die in synthetic nanomaterials. This allows us to use cyanobacteria for the green synthesis of nanomaterials (algal nanotechnology). It can be divided into two types: intracellular synthesis and extracellular synthesis. When synthetic nanomaterials in cells, pigments,enzymes, and polysaccharides use the membrane structure (cell membrane, organelle membrane) to reduce metal ions to realize the conversion of metal ions to metal nanomaterials. Extracellular synthesis of nanomaterials occurs on culture medium or in algae suspension, and the pigments, enzymes, and polysaccharides in the supernatant are used to reduce metal ions to elemental nanoparticles	[71] [72]
Biological drugs	Algae contain a variety of bioactive compounds, including polyunsaturated fatty acids, antioxidants, and unique polysaccharides, which have demonstrated therapeutic potential in various medical applications. These compounds can be extracted and formulated into pharmaceutical products targeting a range of conditions, from cardiovascular diseases to neurodegenerative disorders	[73]

impact. However, challenges persist in achieving low energy consumption and cost-effective, high-efficiency management of cyanobacterial blooms. The ecological toxicity of some nanomaterials, such as silver nanoparticles, which pose risks of toxicity, bioaccumulation, and secondary pollution, remains a significant concern. Xiong et al. [74] researched into the nitrite-responsive hydrogels exemplifies the innovative direction in this field, proposing responsive nanomaterials capable of detecting nutrient levels in water bodies. Such materials could detect nitrogen and phosphorus nutrient salt content in the water and release algaecidal agents under certain conditions, facilitating efficient, low-energy cyanobacterial management. When TN, TP content over boundary conditions, the release of algae control cyanobacteria, not only conducive to our control to the water dose, also can to cyanobacteria adsorption, achieve low energy consumption, high efficiency of cyanobacteria management.

7.2 Policy and regulatory framework for sustainable practices

For effective and sustainable management of HABs, continuous monitoring of water quality and ecology is essential. This involves integrating theories of endogenous nutrient control and exogenous nutrient input to inform and refine cyanobacterial treatment strategies. The establishment of predictive models for water quality and ecology, coupled with real-time remote sensing monitoring, is crucial. These models and data can identify key areas for prevention and control, guiding strategic long-term biological interventions during peak cyanobacteria proliferation seasons (autumn and winter) [75]. Introducing macrophytes and algivorous fish to reduce nitrogen (TN) and phosphorus (TP) concentrations in water bodies can diminish the likelihood of cyanobacterial blooms. Adhering to predictive model results for proactive and responsive control measures is essential in developing a sustainable, ecologically balanced approach to managing HABs.

8 Conclusion

In conclusion, this review elucidates the multifaceted challenges posed by HABs, which are exacerbated by factors such as global climate change and industrial expansion. Our integrated analysis spans the genesis, ecological dynamics, and advanced mitigation strategies for HABs, proposing a multidisciplinary management framework aimed at the systematic control of algal proliferation. Given the complexity of algal ecological dynamics, our review underscores the necessity for strategies that extend beyond conventional methodologies, recommending an integrative application of physical, chemical, and biological controls specifically tailored to diverse environmental conditions. A pivotal aspect of our proposed management strategy is the adoption of advanced technological integrations. Specifically, the employment of artificial intelligence, big data analytics, and remote sensing are essential for the development of accurate predictive models that facilitate enhanced monitoring and management of water quality and ecosystem health. Additionally, this review highlights the transformative potential of algal biomass conversion into value-added products such as ethylene, advanced alcohols, hydroxy acids, and nanomaterials. This not only leverages the photosynthetic and bioremediation capacities of algae but also aligns with the objective of *global carbon neutrality and peaking*, thereby contributing to sustainable environmental stewardship.While the interdisciplinary framework proposed herein offers significant promise, several limitations warrant acknowledgment. The scalability of integrated strategies remains challenging, particularly in large-scale aquatic systems where regional environmental variability (e.g., nutrient profiles, climatic conditions) may necessitate highly tailored implementations. Additionally, the cost-effectiveness of emerging technologies (e.g., nanomaterial-based treatments, AI-driven monitoring systems) remains a barrier to widespread adoption, requiring further optimization for practical application. Long-term ecological risks associated with sustained use of combined methods, such as microbial community disruption or secondary pollutant accumulation, also require more robust monitoring and validation in future studies.Furthermore, we explore the advancement of inorganic nanomaterials in the treatment of cyanobacteria, advocating for the development of innovative photocatalytic materials that are both cost-effective and efficient. The introduction of novel materials such as carbon composites, organic polymers, and selective adsorption technologies is critical for improving the remediation efficacy of cyanobacterial toxins and presents a promising direction for future research. Finally, this review calls for a paradigm shift towards an integrated, interdisciplinary approach in HABs management. This approach should not only aim to mitigate the environmental impacts but also to harness the untapped potential of algae as a sustainable resource. Implementing such holistic strategies is crucial for addressing the growing challenge of HABs in aquatic ecosystems worldwide, thereby paving the way for more sustainable and effective environmental management practices.

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Declarations

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